Preprocessing: comments to paragraph units

```
tx <- list(
  min_tokens = 60L,  # a unit needs at least this many words for the topic model
  max_tokens = 150L,  # a merge may not grow a unit past this; keeps topics apart
  min_sim_quantile = 0.20,  # pairs less similar than this share of all neighbors
never merge
  batch_size = 32L,  # texts per embedding request (the work endpoint allows 37)
  window_n = 300L,  # embedding requests per window
  window_wait = 60L,  # seconds to rest between windows
  use_openrouter = TRUE,  # FALSE at work
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "embed", "COMPASS_EMBED_URL", "COMPASS_EMBED_KEY", "MODEL-FOR-EMBEDDING"))

suppressPackageStartupMessages({
  library(viridis)
  library(tidyverse)  # tidyverse last so its verbs win
})
theme_set(theme_minimal(base_size = 12))
source("../topics_source.R")  # shared functions, one folder up
if (isTRUE(tx$use_openrouter)) source("../llm_openrouter.R")
dir.create(p_out(), showWarnings = FALSE)
```

Corpus

Needs three columns: `id` (comment id), `Year` (integer), `raw_text` (text). The default below is a demonstration corpus that ships with quanteda; at work replace it with one line reading your file.

```
corpus_raw <- tibble(
  id = quanteda::docnames(quanteda::data_corpus_inaugural),
  Year = quanteda::docvars(quanteda::data_corpus_inaugural, "Year"),
  raw_text = as.character(quanteda::data_corpus_inaugural))
# At work:
# corpus_raw <- readr::read_csv(p_in("YOUR_FILE.csv"))
```

```
miss <- setdiff(c("id", "Year", "raw_text"), names(corpus_raw))
if (length(miss)) stop("corpus_raw is missing: ", str_c(miss, collapse = ", "))
stopifnot("id must be unique" = !anyDuplicated(corpus_raw$id),
                  "Year must be whole numbers" = all(corpus_raw$Year ==
round(corpus_raw$Year), na.rm = TRUE))
n0 <- nrow(corpus_raw)
corpus_raw <- corpus_raw |>
```

```
  dplyr::mutate(id = as.character(id)) |>
  dplyr::filter(!is.na(Year), !is.na(raw_text), str_squish(raw_text) != "")
writeLines(glue::glue("Kept {nrow(corpus_raw)} of {n0} comments, \\
  {min(corpus_raw$Year)} to {max(corpus_raw$Year)}."))
```

Kept 60 of 60 comments, 1789 to 2025.

From lines to paragraph units

Most comments here break after every sentence, so splitting at line breaks gives orphan sentences too short for a topic model. Three steps rebuild usable units.

1. **Split and repair.** Text is split at line breaks. A line that follows a line without end punctuation and starts lowercase is a wrapped piece of the same sentence and is joined back.
2. **Merge by meaning.** Every block is embedded. Within a comment, the most similar pair of *neighboring* blocks is merged first, as long as one of them is shorter than `min_tokens`, the result stays within `max_tokens`, and the two are at least as similar as the `min_sim_quantile` point of all neighboring pairs in the corpus. This repeats until no pair qualifies. Reading order is never broken.
3. **Drop what cannot be placed.** A unit still under `min_tokens` had no neighbor similar enough to join without breaking the size ceiling. It is left out of the topic model and written to `dropped_units.csv` so you can read what was lost.

`min_tokens` is enough text for the topic model to read; `max_tokens` is the point past which a unit starts to hold more than one subject. `topics.qmd` refits the model at other settings to show whether that choice drives the topics.

```
blocks <- split_blocks(corpus_raw)
tibble(measure = c("blocks after rejoining wrapped lines", "median words per
block",
                   "blocks under min_tokens", "most blocks in one comment"),
       value = c(nrow(blocks), stats::median(blocks$n_tokens),
                 sum(blocks$n_tokens < tx$min_tokens), max(table(blocks$id))))
|>
  tbl_paged(caption = "Blocks before merging.")
```

Blocks before merging.

measure	value
blocks after rejoining wrapped lines	1871
median words per block	50
blocks under min_tokens	1075
most blocks in one comment	218

Embedding requests carry up to 32 blocks each, packed across comments, so a long comment is spread over several requests. Each distinct text is embedded once and saved; a stopped run picks up where it left off.

```
emb <- embed_all(blocks, p_out("embeddings.parquet"), tx$steps,
tx$use_openrouter,
                 tx$batch_size, tx$window_n, tx$window_wait)
E <- emb_matrix(blocks, emb)
```

```
min_sim <- stats::quantile(adjacent_sims(blocks, E), tx$min_sim_quantile, names
= FALSE)
units <- link_corpus(blocks, E, tx$min_tokens, tx$max_tokens, min_sim)
paragraphs <- dplyr::filter(units, keep)
dropped <- dplyr::filter(units, !keep)

tibble(measure = c("similarity threshold (cosine)", "units formed", "units kept",
                   "units dropped", "words dropped (share)", "comments with
nothing kept",
                   "median words per kept unit", "median blocks per kept unit"),
       value = c(round(min_sim, 3), nrow(units), nrow(paragraphs), nrow(dropped),
                 round(sum(dropped$n_tokens) / sum(units$n_tokens), 3),
                                          dplyr::n_distinct(corpus_raw$id)   -
dplyr::n_distinct(paragraphs$id),
                                           stats::median(paragraphs$n_tokens),
stats::median(paragraphs$n_blocks))) |>
  tbl_paged(caption = "Merging results. If the share of words dropped or the
comments with nothing kept is large, lower min_tokens or min_sim_quantile.")
```

Merging results. If the share of words dropped or the comments with nothing kept is large, lower min_tokens or min_sim_quantile.

measure	value
similarity threshold (cosine)	0.793
units formed	1121.000
units kept	951.000
units dropped	170.000
words dropped (share)	0.030
comments with nothing kept	0.000
median words per kept unit	125.000
median blocks per kept unit	1.000

```
ggplot(paragraphs, aes(n_tokens)) +
  geom_histogram(bins = 40, fill = viridis(1, begin = 0.35)) +
  geom_vline(xintercept = c(tx$min_tokens, tx$max_tokens), linetype = 2) +
  labs(x = "words per unit", y = "units")
```

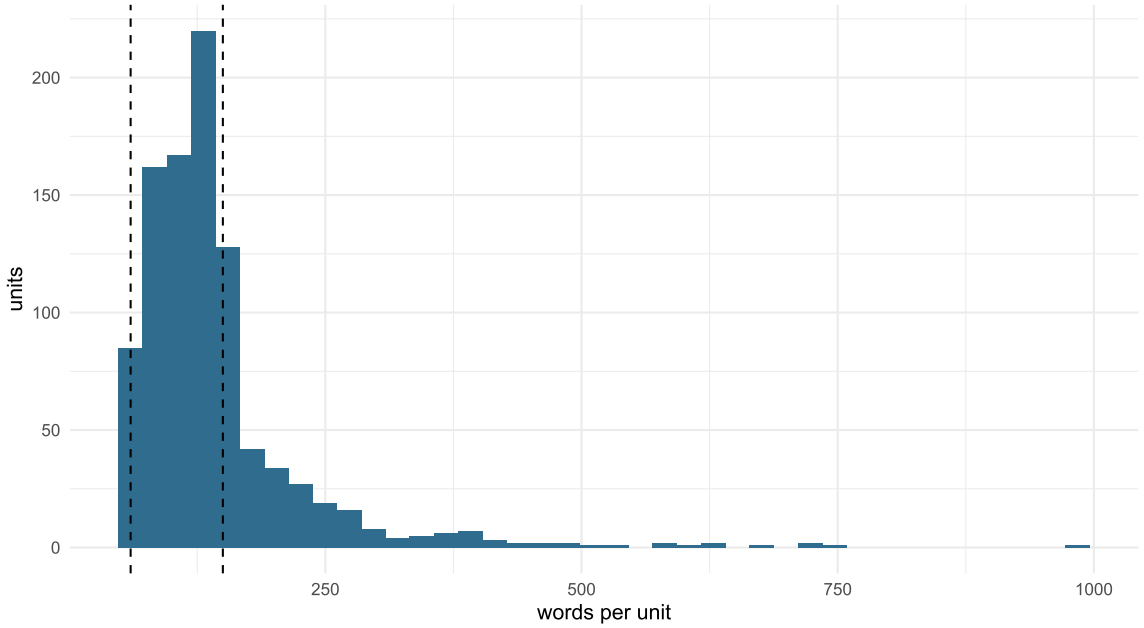


Figure 1: Words per kept unit. Units pile up between min_tokens and max_tokens (dashed); long author paragraphs sit to the right and are never split.

```
units |>
  dplyr::group_by(Year) |>
  dplyr::summarise(dropped = sum(n_tokens[!keep]) / sum(n_tokens)) |>
  ggplot(aes(Year, dropped)) +
  geom_col(fill = viridis(1, begin = 0.6)) +
  labs(x = NULL, y = "share of words dropped")
```

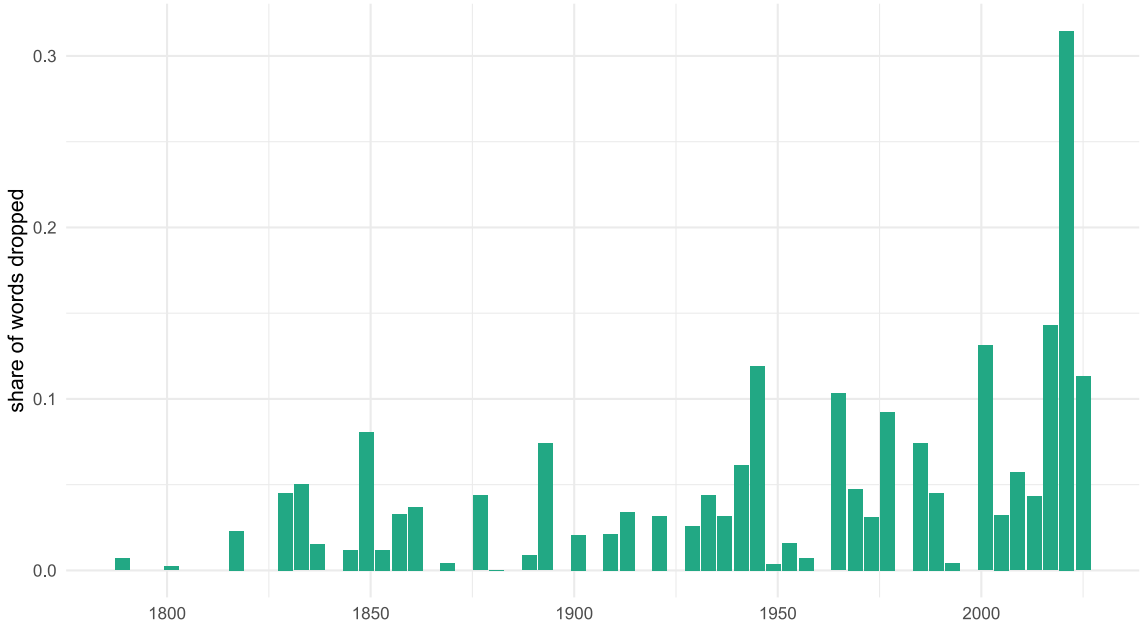


Figure 2: Share of words dropped by year. Dropping is not neutral if it concentrates in some years: those years then rest on longer, better-formed comments.

Handoff

```
readr::write_csv(dplyr::select(paragraphs, para_uid, id, para_id, Year,
n_tokens, n_blocks, text),
                 p_out("paragraphs.csv"))
readr::write_csv(dplyr::select(dropped, id, Year, n_tokens, text),
p_out("dropped_units.csv"))
arrow::write_parquet(blocks, p_out("blocks.parquet"))
arrow::write_parquet(tibble(min_tokens = tx$min_tokens, max_tokens =
tx$max_tokens,
                            min_sim = min_sim), p_out("merge_settings.parquet"))
writeLines(glue::glue("Wrote paragraphs.csv ({nrow(paragraphs)} units), \\
  dropped_units.csv ({nrow(dropped)}), blocks.parquet, merge_settings.parquet,
\\
  embeddings.parquet. Next: topics.qmd."))
```

Wrote paragraphs.csv (951 units), dropped_units.csv (170), blocks.parquet, merge_settings.parquet, embeddings.parquet. Next: topics.qmd.